

5(a)	antigen(s) ; I self- / non-self	1
5(b)	any three from: using bone marrow stem cells 1 (origin from bone marrow so) haematopoietic/results (only) in production of blood cell types ; I multipotent A specialise / differentiate, into red blood cells 2 able to produce genetically identical cells ; 3 (retain ability to) divide (continuously) / carry out mitosis / go through cell cycles ; I reproduce / replicate 4 (so) large numbers (of red blood cells) can be produced ; A can, produce stem cells / self-renew R can differentiate into stem cells 5 can avoid passing on diseases / AW ; from the blood sample 6 less invasive than extraction from bone marrow ; AW ora 7 immobilisation method used to collect the stem cells is, only possible if in blood / easy / easier (than extraction from bone marrow) ; ora 8 AVP ; e.g. can replace blood taken as a sample by a transfusion easily genetically modified can be modified to avoid rejection	3

Question	Answer	Marks
5(c)	any three from: 1 no toxic effects (in recipient) ; 2 should not stimulate an, allergic/immune, response (in the recipient) ; A not rejected 3 does not carry any pathogens / AW ; A no risk of infection / named example A screening (of blood product) not required <i>allow safe to use if mp1, 2 or 3 not gained</i> 4 long shelf life / long-lasting ; A easy to store 5 can transport oxygen ; A idea of improved ability to carry / transport / deliver, oxygen 6 soluble, in blood or idea of able to circulate easily in blood ; 7 used for all blood types / specific to particular blood types ; 8,9 AVP ;; e.g. can carry out other functions of blood fast production / large quantities / mass production acceptable to be used by people because of, ethical / cultural / religious, reasons AW	3